Cover Letter

18 July 2026

Dear Guest Editors and Editors of *Applied Sciences*,

We are pleased to submit the manuscript entitled "Cross-Cohort Benchmarking of Frozen Geneformer and Expression Pseudobulk for Donor-Level SLE Classification" for consideration as an Article in the *Applied Sciences* Special Issue "Research on Computational Biology and Bioinformatics."

Single-cell RNA sequencing measures cells, whereas clinical prediction and biological inference are usually made at the patient level. The central applied question is therefore not only whether a representation separates cases and controls within a dataset, but whether it preserves disease information after the cohort, population, and technical context change. Our study evaluates this question in three public systemic lupus erythematosus cohorts using a strictly donor-level and source-frozen design.

The manuscript makes three contributions. First, it provides reciprocal external-transfer comparisons between mean-pooled frozen Geneformer embeddings and strong expression-pseudobulk baselines, with source-only preprocessing and paired, multiplicity-adjusted statistical testing. Second, it maps representation performance across source-donor size and per-donor cell budgets, converting a model comparison into an operational selection problem. Third, it relates predictive performance to the established interferon program through training-fold residualization and matched-module analyses. The results show that high within-cohort AUC does not identify the most transferable representation and establish source-fitted pseudobulk as an important benchmark for future learned patient-level models.

The manuscript fits the Special Issue because it combines an applied computational-biology question, rigorous benchmarking of bioinformatic representations, and a reproducible workflow addressing a concrete biological and translational problem. The study does not claim clinical diagnostic readiness; its contribution is an externally validated framework for representation selection under cohort shift.

This manuscript, in whole or in part, has not been published previously and is not currently under consideration for publication in another journal. All authors have read and approved the manuscript and agree with its submission to *Applied Sciences*. The authors declare no conflicts of interest. The study uses public, de-identified datasets. Reproducibility materials are available at https://doi.org/10.5281/zenodo.20813922 and https://github.com/LightChainr/rheumlens.

Thank you for your consideration.

Sincerely,

Dan Liu, MD
On behalf of Hongyu Ying, Dandan Yun and Dan Liu
Department of Rheumatology and Immunology
Shanghai Pudong Hospital, Fudan University Pudong Medical Center
Shanghai 201399, China
Email: danliu600@126.com